HAO YU

http://cs-people.bu.edu/haoyu/
Department of Computer Science, Boston University, Boston, MA, 02215
+1-617-372-1630 ⋄ haoyu@bu.edu

EDUCATION

Boston University, Boston, MA, USA *Sep 2019 - Present*
PhD Student in Computer Science
Advisor: Prof. Margrit Betke
GPA: 3.97 / 4.0

Zhejiang University, Hangzhou, Zhejiang, China *Sep 2015 - Jun 2019*
B.E. in Computer Science and B.A. in English Language and Literature
GPA: 3.95 / 4.0

RESEARCH PROJECTS

Leveraging Affect Transfer Learning for Behavior Prediction in an Intelligent Tutoring System January 2021 - May 2021
Supervisor: Prof. Margrit Betke *Graduate reseacher at Boston University*

- Proposed a video-based transfer learning approach for predicting problem outcomes of students working with an intelligent tutoring system (ITS) by analyzing their faces and gestures.
- Created a large labeled dataset of student interactions with an intelligent online math tutor.
- Our model achieved a 50% relative increase in mean F-score over the previous state-of-the-art method on this new dataset.

Measuring and Integrating Facial Expressions and Gaze as Indicators of Engagement and Affect in Tutoring Systems September 2020 - January 2021
Supervisor: Prof. Margrit Betke *Graduate reseacher at Boston University*

- Conducted two studies using computer vision techniques to measure students' engagement and affective states from their head pose and facial expressions, as they use an online tutoring system.
- Presented a Head Pose Tutor, which estimates the real-time head direction of students and responds to potential disengagement, and a Facial Expression-Augmented Teacher Dashboard, that identifies students' affective states and provides enhanced information to teachers.
- Preliminary results on our collected video data indicated promising accuracy in detecting head orientation. A usability study was conducted where the participants rated the system highly in general.

Facial Emotion Analysis for Videos of Presidential Candidates September 2020 - Present
Supervisor: Prof. Margrit Betke *Graduate reseacher at Boston University*

- Collected a large video database of YouTube videos and TV news of 2020 US presidential candidates.
- Annotated the videos with emotion labels using crowd sourcing on Amazon Mechanical Turk.
- Proposed a multi-modal system for facial emotion recognition, which uses pretrained ResNet-18 to extract facial features and uses BERT (Bidirectional Encoder Representations from Transformers) to extract text features. A frame attention module was then adopted for video-based recognition.

Fake Face Video Detection Using Talking Profile January 2019 - April 2019
Supervisor: Prof. Terence Sim *Research Intern at National University of Singapore*

- Proposed a detection algorithm for recognizing fake face videos generated by face swapping using autoencoders and generative adversarial networks (GANs).

- Used talking profile which consists of multiple types of facial and body motions to identify deepfakes.
- Trained support vector machines (SVMs) for classifying real or deepfake videos in DeepfakeTIMIT dataset. Achieved promising classification performance.

Hide-and-Seek: A Data Augmentation Technique Jul 2018 - Sep 2018
Supervisor: Prof. Yong Jae Lee *Research Intern at University of California, Davis*

- Applied Hide-and-Seek, a general purpose data augmentation technique on improving performance of CNN models of Emotion Recognition, Age Estimation, Gender Estimation and Person Re-identification.
- Achieved 1% - 5% relative performance improvement by applying Hide-and-Seek for all tasks. Showcased the advantage of Hide-and-Seek on various visual recognition problems.

PROFESSIONAL ACTIVITIES

Reviewer for IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI) *2019*

TEACHING EXPERIENCE

Introduction to Computer Science II (CS 112), Teaching Fellow, Boston University
Instructors: Prof. Christine Papadakis-Kanaris and Prof. David G. Sullivan *2020 Spring*
Image and Video Computing (CS 585), Teaching Fellow, Boston University
Instructors: Prof. Margrit Betke *2021 Spring*

RELEVANT COURSEWORK

- **Graduate Coursework:** Deep Learning, Machine Learning, Big Data Systems for Data Science, Optimization Theory II, Randomness in Computing.
- **Undergraduate Coursework:** Computer Vision, Computer Graphics, Artificial Intelligence, Information Retrieval, Algorithms.

TECHNICAL STRENGTHS

Computer Languages	Python, C/C++, Java, MATLAB, LaTeX, JavaScript
DL Frameworks	PyTorch, Tensorflow, Caffe, Keras
Toolkits	OpenCV, Scikit-learn, SciPy, Pandas, Numpy, CUDA, MySQL, Hadoop

PUBLICATIONS

[1] **Hao Yu**, Ankit Gupta, Will Lee, Ivon Arroyo, Margrit Betke, Danielle Allesio, Tom Murray, John Magee, and Beverly P. Woolf. ”Measuring and Integrating Facial Expressions and Head Pose as Indicators of Engagement and Affect in Tutoring Systems.” In International Conference on Human-Computer Interaction, pp. 219-233. Springer, Cham, 2021.

[2] Nataniel Ruiz, **Hao Yu**, Mona Jalal, Danielle Allessio, Ajjen Joshi, Tom Murray, Vitaly Ablavsky, John Magee, Jacob Whitehill, Ivon Arroyo, Beverly Woolf, Stan Sclaroff, Margrit Betke. ”Leveraging Affect Transfer Learning for Behavior Prediction in an Intelligent Tutoring System.” In 2021 16th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2021), IEEE, 2021 (accepted).

[3] Kevin Delgado, Juan Manuel Origgi, Tania Hasanpoor, **Hao Yu**, Danielle Allessio, Ivon Arroyo, William Lee, Margrit Betke, Beverly Woolf, and Sarah Adel Bargal. ”Student Engagement Dataset.” In Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 3628-3636. 2021.

[4] Krishna Kumar Singh, **Hao Yu**, Aron Sarmasi, Gautam Pradeep, and Yong Jae Lee. ”Hide-and-seek: A data augmentation technique for weakly-supervised localization and beyond.” arXiv preprint arXiv:1811.02545, 2018.